

Deep Reinforcement Learning for Traffic Light Control

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Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—component, formatting, style, styling, insert.

I. INTRODUCTION

- What are SUMO limitations? - Do we optimize traffic lights only for cars or also bikes, pedestrians, etc. - Do we focus on multiple intersections or just one? - Are we using acyclical or cyclical light switches? May one of them be more realistic than the other? - Do we train on multiple scenarios, like “rush hours” and low traffic conditions, or do we only have normal traffic? - Do we use DQN or are there also other choices like PPO (is PPO able to combine with neural nets?)?

a good starting tutorial on how to generate traffic scenarios can be found here [1].

- at the end, compare normal (static) traffic light with our solution. if possible, the static traffic light control should be decently good. maybe its possible to use a real example?

II. NETWORKS

A. Jönköping

map of sweden downloaded here:
<https://download.geofabrik.de/europe/sweden.html> geojson
polygon object created here: <https://geojson.io/> chosen
geojson found here¹.

then, we used osmium and osmiumfilter to apply the polygon shape to the map of sweden. afterwards, we filter only for highways, except for those of the following types:

- footway
- path
- pedestrian
- steps
- cycleway
- bridleway

finally, we use the SUMO *netconvert* tool to convert the map into a SUMO network.

¹<https://github.com/benbekir/traffic-rl/blob/main/simulations/networks/jkpg/artifacts/jkpg.geojson>



Fig. 1. The filter (in grey) that defines the boundaries of the road network in Jönköping.

1) *Traffic Assignment Zones*: TODO: kick this section out since we dont do anything with this information

Traffic Analysis Zones (TAZs) are spatial units used in transportation modeling and traffic studies to represent areas with similar travel behavior. A TAZ typically models a neighborhood or district, and serves as a fundamental unit for estimating travel demand. Each zone can be associated with socio-economic and demographic data, such as population, employment, and land use. (COPIED FROM [1]).

since we have no information about the actual TAZ's in Jönköping, we create these zones randomly with tools provided by SUMO. there are two options: *gridDistricts*, and *generateBidiDistricts*. lets visualize them:

we choose *gridDistricts* since it creates more realistic traffic assignment zones.

2) *Demand Modeling*: since there is no publicly available data regarding traffic assignment zones and demand in jönköping, we use the SUMO tool *randomTrips* ² with the random seed set to 12 to randomly create trips. we create 3600 individual trips in total where a new trip is spawned into the simulation on every timestep.

3) *Rush Hours and Noise*:

²<https://sumo.dlr.de/docs/Tools/Trip.html#randomtrips>

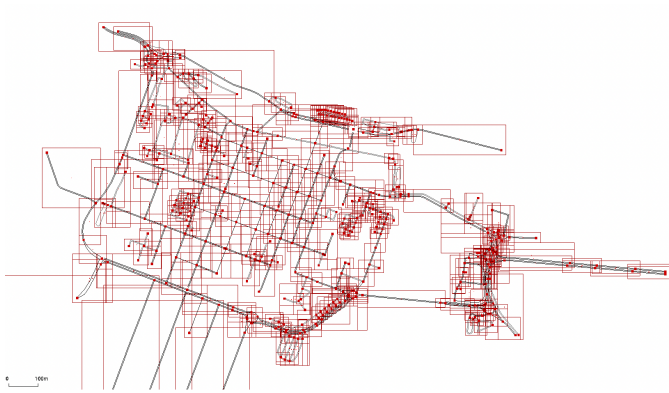


Fig. 2. TAZ created with the generateBidiDistricts tool.

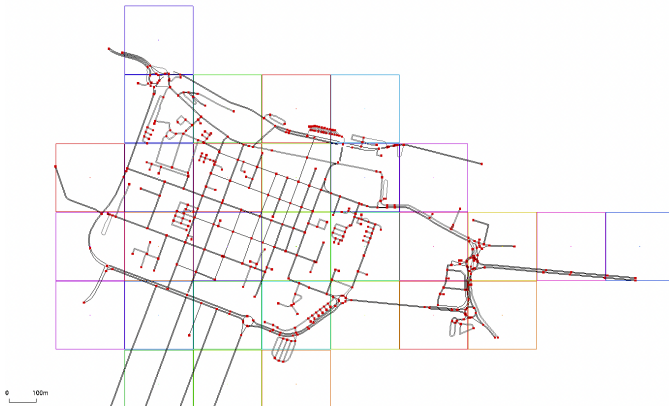


Fig. 3. TAZ created with the gridDistricts tool and with a grid width of 250.

REFERENCES

- [1] D. A. Guastella, E. Montero-Porras, A. Morales-Hernández, and G. Bontempi, "Traffic modeling with sumo: a tutorial," 2025. [Online]. Available: <https://arxiv.org/abs/2304.05982>